Automated Evaluation of Large Language Model Response Concordance with Human Specialist Responses on Physician-to-Physician eConsult Cases

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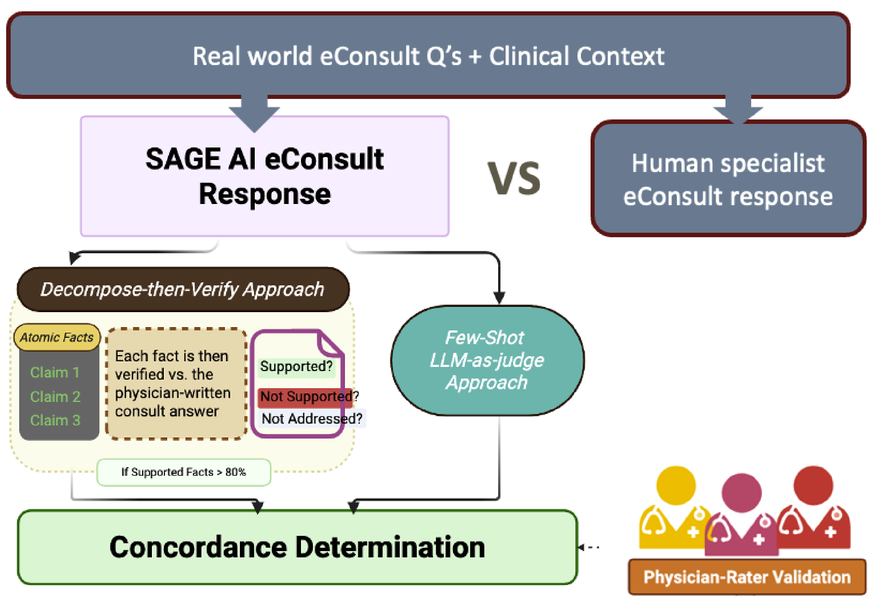
BACKGROUND

- eConsults: A way for PCPs to request specialist input by chart review only.
- Large language models (LLMs) hold promise in reducing such burdens but rigorous evaluation of their performance is lacking and labor-intensive.

Q1: How clinically concordant are AI eConsult responses with human MD responses?
Q2: Are there ways to reliably scale concordance evaluation using AI-assisted methods?

STUDY DESIGN



LLM-as-judge is superior to Decompose-then-Verify (for now) in determining clinical concordance and reaches human rater-level agreement

Check out our pre-prints below!

Concordance Eval *Auto-Template Gen*

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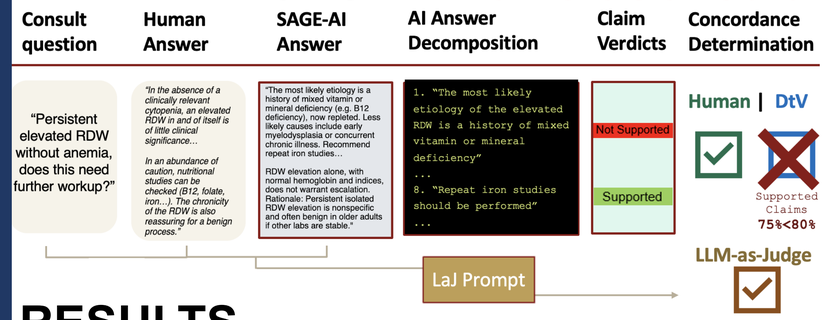
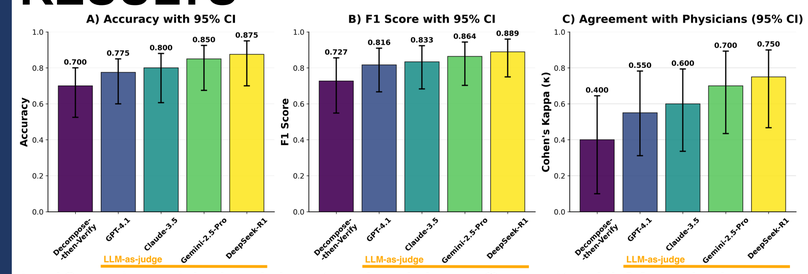
EXAMPLE CASE – Heme eConsult



RESULTS



In 40 real-world physician-to-physician eConsults, LLM-as-judge outperformed decompose-then-verify, achieving human-level concordance assessment with F1-score of 0.89 (95% CI: 0.750, 0.960) and Cohen's kappa of 0.75 (95% CI 0.47,0.90) —comparable to physician inter-rater agreement κ = 0.69-0.90 (95% CI 0.43- 1.0). DeepSeek R1 achieved the highest performance with near-human-level agreement (κ = 0.75, 95% CI 0.47-0.90), comparable to physician inter-rater agreement κ = 0.69-0.90 (95% CI 0.43-1.0).

FUTURE DIRECTIONS

- Establish robust and reliable human concordance metric.
- Larger scale study with remaining 9k+ cases. Further prompt optimization.